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Multi-layer coil component

Abstract

In the multi-layer coil component, the lead conductor is exposed from the end surface of the element body and is connected to the external electrode provided on the end surface. When the lead conductor is led out to the end surface of the element body, the lead area can be easily increased as compared with the case where the lead conductor is led out to the side surface of the element body. Therefore, by connecting the coil and the external electrode via the lead conductor, high connectivity between the coil and the external electrode is achieved.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is based upon and claims the benefit of priority from Japanese Patent Application No. 2021-049717, filed on 24 Mar. 2021, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure relates to a multi-layer coil component.

BACKGROUND

(3) Conventionally, known in the art is a multi-layer coil component in which a coil having a coil axis parallel to a stacking direction is provided in an element body having a stacking structure. Japanese Patent Laid-Open No. 2003-31423 (Patent Document 1) discloses a technique of forming a conductor pattern constituting a coil by using a printing method.

SUMMARY

(4) In the above-described multi-layer coil component according to the conventional art, as shown in FIGS. 12 and 13, coil patterns 21 and 22 constituting a coil 20 and a lead conductor 30A for leading out each of the end portions of the coil 20 to the side surface of an element body 12 (that is,

the surface extending parallel to the stacking direction of the element body 12) are formed by a printing method. Therefore, the lead-out area of the coil 20 is substantially the same as the cross-sectional area of the conductor pattern constituting the lead conductor 30A.

(5) The inventors of the present invention have repeatedly studied the lead-out area of the coil, and have newly found a technique of increasing the lead-out area of the coil to improve the connectivity between the coil and the external electrode.

(6) According to various aspects of the present disclosure, there is provided a multi-layer coil component in which connectivity between a coil and an external electrode is improved.

(7) A multi-layer coil component according to an aspect of the present disclosure comprising, a sintered element body including a plurality of stacked layers and having a pair of end surfaces facing each other in a first direction parallel to a stacking direction of the plurality of layers, a coil provided in the sintered element body and having a coil axis parallel to the first direction, a pair of external electrodes respectively provided on the end surfaces of the sintered element body; and, a pair of lead conductors each provided between the end portion of the coil and the end surface of the sintered element body, the pair of lead conductors being connected electrically to the end portions of the coil and exposed from the end surfaces of the sintered element body to connect to the external electrodes, wherein the lead conductor includes a plurality of lead layers each having a first end portion and a second end portion located closer to the end surface side than the first end portion and displaced from the first end portion when viewed from the first direction, the plurality of lead layers is overlapped in the first direction, and wherein, in the lead layers adjacent to each other in the first direction, the first end portion of one lead layer and the second end portion of the other lead layer overlap each other.

(8) In the multi-layer coil component, the lead conductor connected to the end portion of the coil is exposed from the end surface of the sintered body and connected to the external electrode provided on the end surface. In the case where the lead conductor is led out to the end surface of the sintered element body, the lead area can be easily increased as compared with the case where the lead conductor is led out to the side surface of the element body, and high connectivity between the coil and the external electrode can be achieved.

(9) In the multi-layer coil component according to another aspect, each of the plurality of lead layers has an I-shape when viewed from the first direction, the plurality of lead layers is overlapped in a zigzag manner.

(10) In the multi-layer coil component according to another aspect, a length of the lead conductor in the first direction is 5% or more of a length of the sintered element body in the first direction and is equal to or less than an inner diameter of the coil.

(11) In the multi-layer coil component according to another aspect, the lead conductor includes a first lead layer exposed at the end surface of the sintered element body and a second lead layer overlapping the first lead layer on the coil side, and an area of the first lead layer when viewed from the first direction is larger than an area of the second lead layer.

(12) In the multi-layer coil component according to another aspect, the sintered element body has a print stacking structure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective view showing a multi-layer coil component according to an embodiment.

(2) FIG. 2 is a perspective view showing an inner conductor of the element body shown in FIG. 1.

(3) FIG. 3 is a cross-sectional view taken along line III-III of the multi-layer coil component shown in FIG. 1.

- (4) FIG. 4 is a perspective view showing a stacking structure of the coils shown in FIGS. 2 and 3.
- (5) FIGS. 5A to 5D are views showing each step in manufacturing the coil shown in FIG. 4.
- (6) FIGS. 6A to 6D are views showing each step in manufacturing the coil shown in FIG. 4.
- (7) FIG. 7 is a perspective view showing a stacking structure of the lead conductors shown in FIGS. 2 and 3.
- (8) FIGS. 8A to 8D are views showing each step in manufacturing the lead conductor shown in FIG. 7.
- (9) FIGS. 9A to 9D are views showing each step in manufacturing the lead conductor shown in FIG. 7.
- (10) FIG. 10 is a schematic cross-sectional view showing the lead conductor shown in FIG. 7.
- (11) FIG. 11 is a schematic cross-sectional view showing a stacking structure of lead conductors of different forms.
- (12) FIG. 12 is a perspective view showing an inner conductor of a multi-layer coil component according to the conventional art.

(13) FIG. 13 is a perspective view showing a stacking structure of the coil shown in FIG. 12.

DETAILED DESCRIPTION

(14) Hereinafter, embodiments of the present disclosure will be described with reference to the accompanying drawings. In the description of the drawings, the same or equivalent element is denoted by the same reference numeral, and redundant description is omitted.

(15) A structure of a multi-layer coil component according to an embodiment will be described with reference to FIGS. 1 to 3. As shown in FIG. 1, the multi-layer coil component 10 according to the embodiment includes an element body 12 (sintered element body) and a pair of external electrodes 14A and 14B.

(16) The element body 12 has a substantially rectangular parallelepiped outer shape and includes a pair of end surfaces 12a and 12b facing each other in the extending direction of the element body 12. The element body 12 further includes four side surfaces 12c to 12f extending in the direction in which the end surfaces 12a and 12b face each other and connecting the end surfaces 12a and 12b to each other. In the present embodiment, the side surface 12d is a mounting surface facing the mounting base when the multi-layer coil component 10 is mounted, and the side surface 12c facing the side surface 12d is a top surface when the multi-layer coil component 10 is mounted. When the dimension of the element body 12 in the facing direction of the end surfaces 12a and 12b is a length, the dimension in the facing direction of the side surfaces 12e and 12f is a width, and the dimension in the facing direction of the side surfaces 12c and 12d is a thickness, the dimension of the element body 12 is, for example, 1.6 mm length×0.8 mm width×0.8 mm thickness.

(17) The pair of external electrodes 14A and 14B are provided on the end surfaces 12a and 12b of the element body 12, respectively. In the present embodiment, the external electrode 14A integrally covers the entire region of the end surface 12a and the side surfaces 12c to 12f of the region adjacent to the end surface 12a. Similarly, the external electrode 14B integrally covers the entire region of the end surface 12b and the side surfaces 12c to 12f of the region adjacent to the end surface 12b. Each of the external electrodes 14A and 14B includes one or more electrode layers. For example, a metallic material such as Ag may be used as an electrode material constituting each of the external electrodes 14A and 14B.

(18) The element body 12 has a structure in which an internal conductor 18 is provided inside a magnetic body 16. The element body 12 has a stacking structure. The magnetic body 16 has a stacking structure in which a plurality of magnetic layers 16L and 16R described later are stacked in the direction in which the end surfaces 12a and 12b face each other. In the following description, the facing direction of the end surfaces 12a and 12b is also referred to as a stacking direction or a first direction of the element body 12.

(19) The magnetic body 16 is made of a magnetic material such as ferrite. The magnetic body 16 is obtained by stacking and sintering a plurality of magnetic pastes (for example, ferrite pastes) to be

the magnetic body layers **16L** and **16R**. That is, the element body **12** has a print stacking structure and is a sintered element body, in which the magnetic layers **16L** and **16R** on which the magnetic paste is printed are stacked and sintered. The number of magnetic layers **16L** and **16R** constituting the element body **12** is, for example, 30 layers. The thickness of each of the magnetic layers **16L** and **16R** is, for example, 30 μm . In the actual element body **12**, the plurality of magnetic layers **16L** and **16R** are integrated such that boundaries between the layers are not visible.

(20) The inner conductor **18** includes one coil **20** and a pair of lead conductors **30**. Each of the coil **20** and the lead conductor **30** of the inner conductor **18** has a stacking structure in the stacking direction of the element body **12**. As shown in FIG. 2, in the present embodiment, the coil **20** has a cylindrical outer shape.

(21) As shown in FIG. 3, the coil **20** has a coil axis Z parallel to the stacking direction of the element body **12** and is wound around the coil axis Z. In the present embodiment, the length of the coil **20** in the stacking direction of the element body **12** is 1.3 mm. In the stacking direction of the element body **12**, the length of the coil **20** can be designed to be in a range of 50 to 90% of the length of the element body **12**. In the present embodiment, the inner diameter of the coil **20** is 0.20 to 0.40 mm, for example, 0.35 mm.

(22) The coil **20** includes a plurality of coil layers **21** and **22**. The coil layers **21** and **22** constituting the coil **20** are made of a conductive material containing a metal such as Ag. The coil **20** is formed by a printing method. Specifically, the coil **20** is obtained by applying a conductive paste (for example, Ag paste) to be the coil layers **21** and **22** on a magnetic paste to be the magnetic layers **16L** and **16R** and sintering the conductive paste. The thickness of each of the coil layers **21** and **22** is, for example, 20 μm .

(23) FIG. 4 is an exploded perspective view showing the stacking structure of the portion of the coil **20** of the element body **12**. As shown in FIG. 4, in the portion of the coil **20** of the inner conductor **18**, the coil layers **21** and **22** constituting the coil **20** and the magnetic layers **16L** and **16R** constituting the element body **12** are alternately stacked.

(24) Each of the coil layers **21** and **22** has a C-shape (or a U-shape) when viewed from the stacking direction of the element body **12**. When viewed from the stacking direction of the element body **12**, the coil layer **21** and the coil layer **22** have a relationship of point symmetry or rotational symmetry with respect to the coil axis Z. The coil layer **21** and the coil layer **22** constitute $\frac{3}{4}$ turns of the coil **20**. The coil layers **21** and the coil layers **22** are alternately arranged in the stacking direction of the element body **12**. The coil layer **21** and the coil layer **22**, which are adjacent to each other in the stacking direction of the element body **12**, are joined to each other with their end portions overlapping each other, and constitute one turn of the coil **20** surrounding the coil axis Z.

(25) FIGS. 5A to 5D and 6A to 6D are views showing a procedure for forming the coil **20** by the printing method.

(26) As shown in FIG. 5A, first, the conductive paste for forming the coil layer **22** is printed on the magnetic layer served as a base. The coil layer **22** has a first end portion **22a** on the upper side of the left half surface and a second end portion **22b** on the lower side of the right half surface.

(27) Next, as shown in FIG. 5B, the magnetic paste for forming the magnetic layer **16L** is printed on the left half surface. Accordingly, only the first end portion **22a** of the end portions **22a** and **22b** of the coil layer **22** is covered with the magnetic paste.

(28) Then, as shown in FIG. 5C, the conductive paste to be the coil layer **21** is printed. The coil layer **21** has a first end portion **21a** on the lower side of the right half surface and a second end portion **21b** on the upper side of the left half surface. The first end portion **21a** of the coil layer **21** located on the right half surface overlaps and is joined to the second end portion **22b** of the coil layer **22** exposed from the magnetic layer **16L**. The second end portion **21b** of the coil layer **21** located on the left half surface is provided on the magnetic layer **16L**. Therefore, the first end portion **21a** and the second end portion **21b** of the coil layer **21** are at different height positions in the stacking direction of the element body **12**. More specifically, the second end portion **21b** is

located closer to the end surface **12a** (closer to the viewer of FIGS. 5A to 5D) than the first end portion **21a**.

(29) Further, as shown in FIG. 5D, the magnetic paste to be the magnetic layer **16R** is printed on the right half surface. Accordingly, only the first end portion **21a** of the end portions **21a** and **21b** of the coil layer **21** is covered with the magnetic paste.

(30) Subsequently, as shown in FIG. 6A, the conductive paste to be the coil layer **22** is printed again. The first end portion **22a** of the coil layer **22** positioned on the left half surface overlaps and is joined to the second end portion **21b** of the coil layer **21** exposed from the magnetic layer **16R**. The second end portion **22b** of the coil layer **22** located on the right half surface is provided on the magnetic layer **16R**. Therefore, the first end portion **22a** and the second end portion **22b** of the coil layer **22** are at different height positions in the stacking direction of the element body **12**. More specifically, the second end portion **22b** is located closer to the end surface **12a** (closer to the viewer of FIGS. 6A to 6D) than the first end portion **22a**.

(31) Then, as shown in FIG. 6B, the magnetic paste for forming the magnetic layer **16L** is printed again on the left half surface. Accordingly, only the first end portion **22a** of the end portions **22a** and **22b** of the coil layer **22** is covered with the magnetic paste.

(32) Further, as shown in FIG. 6C, the conductive paste to be the coil layer **21A** is printed. The coil layer **21A** has substantially the same configuration as the coil layer **21** described above, and is different from the coil layer **21** only in the position of the second end portion **21b**. The position of the second end portion **21b** of the coil layer **21A** is adjusted for connection with the lead conductor **30** described later. The second end portion **21b** of the coil layer **21A** is located substantially in the middle of the left half surface (substantially in the middle in the vertical direction in FIG. 6C). The first end portion **21a** of the coil layer **21A** overlaps and is joined to the second end portion **22b** of the coil layer **22** exposed from the magnetic layer **16L**. The second end portion **21b** of the coil layer **21A** is provided on the magnetic layer **16L**. In the coil layer **21A**, similarly to the coil layer **21**, the first end portion **21a** and the second end portion **21b** are located at different height positions in the stacking direction of the element body **12**.

(33) Thereafter, as shown in FIG. 6D, the magnetic paste to be the magnetic layer **16R** is printed again on the right half surface. Accordingly, only the first end portion **21a** of the end portions **21a** and **21b** of the coil layer **21A** is covered with the magnetic paste.

(34) One turn of the coil **20** is formed by the print stacking of the coil layer **21** of FIG. 5C, the magnetic layer **16R** of FIG. 5D, the coil layer **22** of FIG. 6A, and the magnetic layer **16L** of FIG. 6B. By setting these as one set and repeating a plurality of sets, the coil **20** constituted by the plurality of turns is formed.

(35) As shown in FIG. 3, the pair of lead conductors **30** are provided between the coil **20** and the end surfaces **12a** and **12b** of the element body **12**, respectively. Each of the lead conductors **30** is electrically connected to the end of the coil **20**, and is exposed from the end surfaces **12a** and **12b** of the element body **12** to be connected to the external electrodes **14A** and **14B**. In the present embodiment, the length of the lead conductor **30** in the stacking direction of the element body **12** is 0.15 mm. In the stacking direction of the element body **12**, the length of the lead conductor **30** can be designed to be in the range of 5 to 25% of the length of the element body **12**.

(36) FIG. 7 is an exploded perspective view showing the stacking structure of the portion of the lead conductor **30** of the element body **12**. As shown in FIG. 7, the lead conductor **30** includes an exposed layer **31** exposed from the end surfaces **12a** and **12b** and a plurality of lead layers **32**, which are stacked. The exposed layer **31** and the lead layer **32** constituting each of the lead conductors **30** are made of a conductive material containing a metal such as Ag. Each of the lead conductor **30** is formed by a printing method similarly to the coil **20**. Specifically, each of the lead conductors **30** is obtained by applying a conductive paste (for example, Ag paste) to be the exposed layer **31** and the lead layer **32** onto a magnetic paste to be the magnetic layers **16L** and **16R** and sintering the conductive paste. The thickness of each of the exposed layer **31** and the lead layer **32**

is, for example, 20 μm .

(37) In the portion of the lead conductor **30** of the internal conductor **18**, the lead layers **32** constituting the lead conductor **30** and the magnetic layers **16L** and **16R** constituting the element body **12** are alternately stacked. The exposed layer **31** is provided on the uppermost lead layer **32**.

(38) The exposed layer **31** has a circular shape when viewed from the stacking direction of the element body **12**, and is located on the coil axis Z of the coil **20**. Each of the lead layers **32** has an I-shape when viewed from the stacking direction of the element body **12**, and is located on the coil axis Z of the coil **20**. In this embodiment, the lead layers **32** have the same dimensions. The plurality of lead layers **32** entirely overlap each other when viewed from the stacking direction of the element body **12**. The plurality of lead layers **32** are arranged in the stacking direction of the element body **12**, and the adjacent lead layers **32** are joined with the end portions **32a** and **32b** overlapping each other.

(39) When viewed from the stacking direction of the element body **12**, the area of the exposed layer **31** is designed to be larger than the area of the lead layer **32**. In particular, the exposed layer **31** is designed to have a larger area than the uppermost lead layer **32**.

(40) FIGS. **8A** to **8D** and **9A** to **9D** are views showing a procedure for forming the lead conductor **30** by a printing method.

(41) First, as shown in FIG. **8A**, the conductive paste to be the lead layer **32** is printed on the coil layer **21A** shown in FIG. **6D**. In FIGS. **8A** to **8D** and **9A** to **9D**, the I-shaped lead layer **32** extends in the left-right direction and has a left end portion **32a** and a right end portion **32b**. The left end portion **32a** of the lead layer **32** overlaps and is joined to the second end portion **21b** of the coil layer **21A** exposed from the magnetic layer **16R**. The right end portion **32b** of the lead layer **32** is provided on the magnetic layer **16R**. In the lead layer **32**, the left end portion **32a** and the right end portion **32b** are at different height positions in the stacking direction of the element body **12**. More specifically, the right end portion **32b** is located closer to the end surface **12a** (closer to the viewer of FIGS. **8A** to **8D**) than the left end portion **32a**.

(42) Next, as shown in FIG. **8B**, the magnetic paste for forming the magnetic layer **16L** is printed on the left half surface. Accordingly, only the left end portion **32a** of the end portions **32a** and **32b** of the lead layer **32** is covered with the magnetic paste.

(43) Then, as shown in FIG. **8C**, the conductive paste to be the lead layer **32** is printed again. The right end portion **32b** of the lead layer **32** overlaps and is joined to the right end portion **32b** of the lead layer **32** exposed from the magnetic layer **16L**. The left end portion **32a** of the lead layer **32** is provided on the magnetic layer **16L**. Therefore, the left end portion **32a** and the right end portion **32b** of the lead layer **32** are at different height positions in the stacking direction of the element body **12**. More specifically, the left end portion **32a** located on the magnetic layer **16L** is located closer to the end surface **12a** (closer to the viewer of FIGS. **8A** to **8D**) than the right end portion **32b**.

(44) Further, as shown in FIG. **8D**, the magnetic paste for forming the magnetic layer **16R** is printed on the right half surface. Accordingly, only the right end portion **32b** of the end portions **32a** and **32b** of the lead layer **32** is covered with the magnetic paste.

(45) Subsequently, as shown in FIG. **9A**, the conductive paste to be the lead layer **32** is printed again. The left end portion **32a** of the lead layer **32** overlaps and is joined to the left end portion **32a** of the lead layer **32** exposed from the magnetic layer **16R**. The right end portion **32b** of the lead layer **32** is provided on the magnetic layer **16R**. Therefore, the right end portion **32b** located on the magnetic layer **16R** is located closer to the end surface **12a** (closer to the viewer of FIGS. **9A** to **9D**) than the left end portion **32a**.

(46) Then, as shown in FIG. **9B**, the magnetic paste for forming the magnetic layer **16L** is printed again on the left half surface. Accordingly, only the left end portion **32a** of the end portions **32a** and **32b** of the lead layer **32** is covered with the magnetic paste.

(47) Further, as shown in FIG. **9C**, the conductive paste to be the lead layer **32** is printed again. The

right end portion **32b** of the lead layer **32** overlaps and is joined to the right end portion **32b** of the lead layer **32** exposed from the magnetic layer **16L**. The left end portion **32a** of the lead layer **32** is provided on the magnetic layer **16L**. Therefore, the left end portion **32a** located on the magnetic layer **16L** is located closer to the end surface **12a** (closer to the viewer of FIGS. **9A** to **9D**) than the right end portion **32b**.

(48) The pair of lead layers **32** jointed each other is formed by print stacking of the lead layer **32** of FIG. **8C**, the magnetic layer **16R** of FIG. **8D**, the lead layer **32** of FIG. **9A**, and the magnetic layer **16L** of FIG. **9B**. By setting these as one set and repeating a plurality of sets, it is possible to form the lead conductor **30** including some pairs of the lead layers **32**.

(49) Thereafter, as shown in FIG. **9D**, the conductive paste to be the exposed layer **31** is printed. The periphery of the exposed layer **31** may be filled with the magnetic paste.

(50) FIG. **10** is a cross-sectional view showing the lead conductor **30** exposed at the end surface **12a** of the element body **12**. The lead conductor **30** on the end surface **12b** of the element body **12** is the same as the lead conductor **30** on the end surface **12a** of the element body **12**, and thus the description thereof is omitted.

(51) As shown in FIG. **10**, the plurality of lead layers **32** are interposed between the magnetic layers **16R** and the magnetic layers **16L** that are alternately stacked. Therefore, the left end portion **32a** and the right end portion **32b** of each of the lead layers **32** are at different height positions with respect to the stacking direction of the element body **12**, and the plurality of lead layers **32** overlap each other in a zigzag manner. In the lead conductor **30**, as indicated by arrows in FIG. **10**, the current path is also zigzag. In FIG. **10**, the plurality of lead layers **32** are shown to overlap in a bellows shape. The plurality of lead layers **32** may continuously overlap in a Z-shape.

(52) As described above, the multi-layer coil component **10** includes the plurality of stacked magnetic layers **16R** and **16L**, the element body **12** (sintered element body) having the pair of end surfaces **12a** and **12b** facing each other in the first direction parallel to the stacking direction of the plurality of magnetic layers **16R** and **16L**, the coil **20** provided in the element body **12** and having the coil axis **Z** parallel to the first direction, and the pair of external electrodes **14A** and **14B** provided on the end surfaces **12a** and **20b** of the element body **12**, and the pair of lead conductors **30** each provided between the end portion of the coil **20** and the end surfaces **12a** and **12b** of the element body **12**, the pair of lead conductors **30** being connected electrically to the end portions of the coil **20** and exposed from the end surfaces **12a** and **12b** of the element body **12** to connect to the external electrodes **14A** and **14B**. Each of the lead conductors **30** includes the plurality of lead layers **32**, and each of the lead layers **32** has the left end portion **32a** and the right end portion **32b**. The left end portion **32a** and the right end portion **32b** are located at different height positions in the first direction, and are displaced from each other when viewed from the first direction. In the lead layers **32** adjacent to each other in the first direction, the end portions **32a** and **32b** (second end portions) positioned on the upper side in the stacking direction of one lead layer **32** and the end portions **32a** and **32b** (first end portions) positioned on the lower side in the stacking direction of the other lead layer **32** overlap each other.

(53) In the multi-layer coil component **10**, the lead conductor **30** is exposed from the end surfaces **12a** and **12b** of the element body **12** and is connected to the external electrodes **14A** and **14B** provided on the end surfaces **12a** and **12b**. In the case where the lead conductors **30** are led out to the end surfaces **12a** and **12b** of the element body **12**, the lead area can be easily increased compared to the case where the lead conductors are led out to the side surfaces **12c** to **12f** of the element body **12**. For example, the exposed layer **31** of the lead conductor **30** exposed from the end surfaces **12a** and **12b** of the element body **12** can easily increase the area when viewed from the stacking direction of the element body **12** (that is, the lead area), and the area can be increased with respect to the cross-sectional area of the lead layer **32** or the coil layers **21** and **22**. Therefore, by connecting the coil **20** and the external electrodes **14A** and **14B** via the lead conductor **30**, high connectivity between the coil **20** and the external electrodes **14A** and **14B** is achieved.

(54) The lead conductor **30** of the multi-layer coil component **10** has a reduced electrode volume. For comparison, FIG. **11** shows a lead conductor **30A** including a plurality of via conductors **33**. Each of the via conductors **33** penetrates each of the magnetic layers **16A** constituting the element body **12**. The plurality of via conductors **33** are linearly arranged along the stacking direction of the element body **12**. In the lead conductor **30A**, the current path is also linear as indicated by the arrow in FIG. **11**. In the lead conductor **30A** shown in FIG. **11**, each of the via conductors **33** is provided so as to fill the through hole of the magnetic layer **16A**, and thus each via conductor **33** has a relatively large volume. Therefore, the electrode volume of the lead conductor **30A** assembled from the plurality of via conductors **33** is increased. In this case, cracks are likely to occur due to the difference in shrinkage between the via conductor **33** and the magnetic layer **16A** during sintering. On the other hand, the lead conductor **30** of the multi-layer coil component **10** has a relatively small volume because the lead layer **32** does not penetrate the magnetic layers **16R** and **16L**. Therefore, cracks are prevented in the multi-layer coil component **10**.

(55) In the multi-layer coil component **10**, the lengths of the lead conductors **30** in the first direction are designed to be 5% or more of the lengths of the element body **12** in the first direction and to be equal to or less than the inner diameter of the coil **20**.

(56) Furthermore, in the multi-layer coil component **10**, the lead conductor **30** includes the exposed layer **31** (first lead layer) exposed at the end surfaces **12a** and **12b** of the element body **12** and the uppermost lead layer **32** (second lead layer) overlapping the exposed layer **31** on the coil **20** side, and the area of the exposed layer **31** when viewed from the first direction is designed to be larger than the area of the uppermost lead layer **32**.

(57) Although the embodiments of the present disclosure have been described above, the present disclosure is not necessarily limited to the above-described embodiments, and various modifications can be made without departing from the gist thereof. For example, the exposed layer **31** of the lead conductor **30** is not limited to one layer, and may be formed of a plurality of layers. The lead conductor **30** may include a plurality of lead layers **32** without including the exposed layer **31**. The number of the pair of the lead layers **32** of the lead conductor **30** may be one or more. The coil **20** may have a circular ring shape, a rectangular ring shape, or an elliptical ring shape when viewed from the stacking direction of the element body **12**.

Claims

1. A multi-layer coil component comprising: a sintered element body including a plurality of stacked layers and having a pair of end surfaces including a first end surface and a second end surface facing each other in a first direction parallel to a stacking direction of the plurality of layers; a coil provided in the sintered element body and having a coil axis parallel to the first direction, the coil having end portions; a pair of external electrodes respectively provided on the end surfaces of the sintered element body; and, a pair of lead conductors each provided between one of the end portions of the coil and one of the end surfaces of the sintered element body, the pair of lead conductors being connected electrically to the end portions of the coil and exposed from the end surfaces of the sintered element body to connect to the external electrodes, wherein the pair of lead conductors includes a first lead conductor that includes a plurality of lead layers each having a first end portion and a second end portion located closer to the first end surface side than the first end portion and displaced from the first end portion when viewed from the first direction, the plurality of lead layers is overlapped in the first direction, wherein, in the lead layers adjacent to each other in the first direction, the first end portion of one lead layer and the second end portion of the other lead layer overlap each other, and wherein the first lead conductor includes a first lead layer exposed at the first end surface of the sintered element body and a second lead layer overlapping the first lead layer on the coil side, and an area of the first lead layer when viewed from the first direction is larger than an area of the second lead layer.

2. The multi-layer coil component according to claim 1, wherein each of the plurality of lead layers has an I-shape when viewed from the first direction, the plurality of lead layers is overlapped in a zigzag manner.
 3. The multi-layer coil component according to claim 1, wherein a length of the first lead conductor in the first direction is 5% or more of a length of the sintered element body in the first direction and is equal to or less than an inner diameter of the coil.
 4. The multi-layer coil component according to claim 1, the sintered element body has a print stacking structure.
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